

Phase 4: Transition — Phase Overview

Resilio — Mobile Mental Health Platform for Nepal | Milan Shrestha | ID: 23057135 | 04_Transition/README.md

RUP Phase: Transition | **Week:** 20 | **Dates:** 19 March – 30 April 2025

Phase Objective

Deploy the complete system to production (Vercel + Railway), conduct system testing and user acceptance testing, gather post-survey data, obtain client approval letters, and finalise the FYP report for submission.

Documents in This Folder

File	Contents	Report Reference
`Testing_Strategy.md`	Three-level testing approach (unit/system/UAT); test design rationale (equivalence partitioning, BVA); coverage table; UAT protocol; traceability matrix mapping all 24 FRs to test cases; defects log (5 bugs)	Chapter 4; Appendix K
`Deployment.md`	Vercel deployment of Node.js/Express backend; actual `vercel.json`; Socket.IO constraint and Railway solution; CORS config; all 14 environment variables; Flutter `ApiEndpoints.dart` pattern; deployment checklist	Chapter 3, Section 3.6

Transition Outcomes

Activity	Result
Unit tests (40 cases)	40/40 passed — 100%
System tests (40 cases)	40/40 passed — 100%
UAT (25 cases, 7 participants)	24 passed, 1 partial — 99.05% overall
Post-survey (132 responses)	All 10 outcome metrics positive
Vercel production deployment	Live at `https://resilio-api.vercel.app`
Socket.IO signalling deployment	Live at `https://resilio-signalling.up.railway.app`
Client approval letters	Obtained from Ms. Sushana Karki and Dr. Dibyandra Singh

Bugs Resolved Before Submission

Bug ID	Description	Resolution
BUG-001	Double-booking race condition on concurrent slot selection	PostgreSQL row-level lock on appointments insert
BUG-002	eSewa verification failing — type mismatch on transaction ID	Coerced both sides to string before comparison
BUG-003	WebRTC ICE failure on 4G networks	Added TURN server fallback; increased ICE timeout to 15s
BUG-004	PHQ-9 discoverability — 3 min vs 2 min target (TC-UAT011 partial)	Promoted questionnaire card to home dashboard
BUG-005	SuperTokens redirect URL mismatch in production	Updated `WEBSITE_DOMAIN` env var to Vercel production URL

Production Deployment Architecture

```
Flutter App (Android APK)
  ↓ REST calls
Vercel (https://resilio-api.vercel.app)
  Node.js / Express – src/index.js
  ↓
Supabase (PostgreSQL) – managed cloud

Flutter App
  ↓ WebSocket
Railway (https://resilio-signalling.up.railway.app)
  Socket.IO signalling server – src/services/webrtc-socket.js
  ↓
WebRTC peer connection (P2P audio/video)
```